Minh Le

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SUMMARY

Second-year mechanical engineering student at UBC drawn to the mechanical side of sustainable technology, who enjoys bridging theory with real, practical builds. Seeking a summer engineering internship or co-op.

EDUCATION

University of British Columbia, Vancouver — BASc, Mechanical Engineering — Vancouver, BC

Second Year · Expected graduation: May 2029 · Cumulative GPA: 4.00/4.33 (A) · Dean's List

PROJECTS

Two-Speed Hand-Crank Generator — Personal project

Hand-cranked, 3D-printed axial-flux generator · SolidWorks, 3D printing, machining, Arduino

- Designed and built a hand-crank generator that sets its output voltage through gear selection instead of regulator electronics, charging a phone from hand power alone.
- Built a two-speed gearbox with a constant-mesh dog clutch that shifts under load; proved with logged data that each gear selects a distinct voltage (high gear ~1.9× low), modeled in SolidWorks and printed in PLA with hand-machined shafts.
- Characterized the hand-wound axial-flux generator with an Arduino, INA219, and Hall sensor; measured drivetrain efficiency up to ~48%.

Leader-Follower Robotic Arm — Personal project

4-DOF arm with custom PCB and firmware · SolidWorks, KiCad, Arduino, 3D printing

- Designed and built a 4-DOF arm that mirrors a hand-held twin in real time, with record and playback.
- Designed a custom Arduino shield PCB in KiCad, manufactured by JLCPCB.
- Wrote three-mode firmware with sensor smoothing; fixed voltage dips and swapped to metal-gear servos for load.

Rainwater Harvester: System Simulation & Design — APSC 101 · 6-person team

5-year spreadsheet model of an off-grid water system · Excel, fluid mechanics, energy modeling

- Built the topography model and a 5-year daily water-balance simulation tracking system reliability.
- Modeled on-demand flow and pump-to-storage iteratively from pipe friction, elevation, and pump curves.
- Placed 2nd of 10 in studio at 63.66% satisfaction on the coordinators' withheld-weather simulation.

TECHNICAL SKILLS

- **CAD & Design:** SolidWorks (modeling, assemblies, print-ready parts); technical drawing (orthographic, isometric)
- **Electronics:** PCB design in KiCad (schematic, layout, IPC-2221 trace sizing); Arduino; servo control; basic circuits
- **Programming:** Embedded C/C++ (Arduino); Python; MATLAB (coursework)
- **Fabrication:** 3D printing (PLA); machining; sheet-metal fabrication
- **Simulation:** Excel-based system modeling (fluids, energy, multi-year reliability)

EXPERIENCE

Club Volleyball Coach — Volleydome — Calgary, AB · May–Aug 2026, Jan 2024 – Jun 2025

- Planned and led practices to build athlete skills, teamwork, and confidence.
- Gave individual feedback and communicated with athletes, parents, and staff.

Volunteer — Volleydome — Calgary, AB · Mar 2023 – Dec 2023

- Supported event setup and logistics and assisted participants during community events.

ADDITIONAL

Languages: English · Availability: May–August (full-time) · Driver's licence: Class 5 (Alberta)